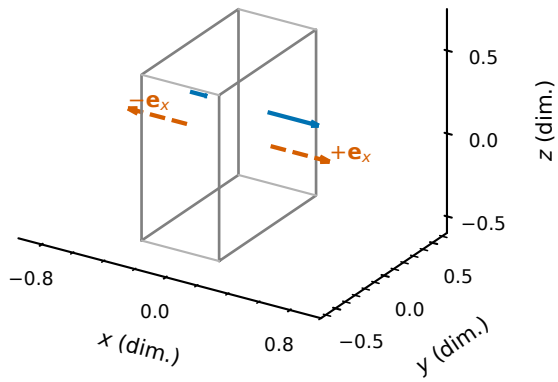
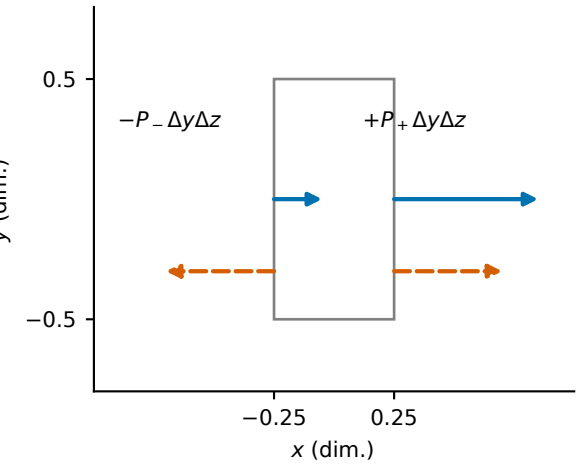
A



Flux: $\mathbf{F} \cdot \mathbf{n}\, dS$; opposite normals

B



$P = 1 + 2x$: $\quad P_+ - P_- = 2\Delta x$